

Hongye Hou

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Education

HKUST-gz University, AI4City Lab, *PhD Candidate* Aug. 2026 - Now

- Research Interests: 3D Generation for point cloud, indoor scene, Lod2 buildings; 3D Point Cloud Detailed Representation and Completion; AI-Agent/VLM fine-tuning for 3D Understanding; Solid industrial internship background with hands-on experience.

Xi'an Jiaotong University, Software Engineering, *Master* Sep. 2023 - Jun. 2026

- GPA: 3.64/4; Ranked top 5%, Recipient of the prestigious XJTU Scholarship.

Northeastern University, Information Management and Systems, *Bachelor* Sep. 2019 - Jun. 2023

- GPA: 3.92/5; Ranked top 5%, Courses: Machine Learning (96), High-Dimensional Data Analysis (90), Linear Algebra (93); CET-4 (541), CET-6 (511), Ielts 6.5 overall (both > 6.0).

First Author and Co-1st Publications

[1] Dig into Detailed Structures: Key Context Encoding and Semantic-based Decoding for Point Cloud Completion. **ACM Multimedia 2024 CCF-A**. **Hou Hongye**, Gao Xuehao, Liu Zhan, Yang Yang[†]

[2] CitySTAR: Structured and Topology-Aware Reasoning for Open-Vocabulary Urban 3D Grounding. **NeurIPS 2026 CCF-A**. Shuai Zhang*, **Hongye Hou***, Qinghe Liu, Zhuoxiao Li, Dongli Wu, Yuan Liu, Wufan Zhao[†]

[3]Benefit from reference: Retrieval-augmented cross-modal point cloud completion, **ASOC((JCR Q1; IF: 7.8))**. **Hongye Hou**, Yang Yang[†], Zhan Liu, Fangyu Du

[4] Hou Hongye et al., "Bri3D: Bridging the Object-Scene Gap in Single-Image 3D Generation." Submitted.

[1] <https://dl.acm.org/doi/abs/10.1145/3664647.3680565>, [3] <http://arxiv.org/abs/2507.14485>

Industry Internship Experience

Alibaba - TaoBao Meta Technology Lab, 3DAIGC Algorithm Engineer Dec. 2024 - Oct. 2025

- Focusing on compositional 3D generation, designed a two-stage generative scheme to solve 3D part and scene generation problem. In the first stage, a MASK image is injected as a condition to generate voxels with labels. Secondly, using a normal map and introducing part-attention to generates high-resolution Meshes.
- Proposed a 3D Mesh data synthesis and quality assessment pipeline, which, through massive generation of 2D image-> 3D data and iterative filtering, distills high-quality core training assets from open-source models.
- Involved in a fashion product recommendation system utilizing multimodal large models.

Huawei Technologies Co., Ltd., Code LLM Algorithm Engineer Jul. 2024 - Oct. 2024

- **Core Project (Data Quality)**: Designed algorithms to enhance code data quality by building a scoring model to classify three types of language data. Implemented strategies including high-quality data synthesis, rigorous pre-training data cleaning, variable renaming, and comment augmentation/back-translation, resulting in a **20 improvement** in code generation capabilities.
- **Automated Evaluation**: Developed an automated evaluation pipeline for Code-LLMs, significantly improving assessment efficiency and iteration speed.
- **Data Generation Safety**: Optimized prompts for efficient large-scale data generation and constructed a comprehensive dataset of malicious attack prompts for robust LLM safety testing.

MechMind Robotics (Beijing), Machine Vision Algorithm Engineer Nov. 2022 - Mar. 2023

- Responsible for improving clustering algorithms in object grasping processes, introduced innovative region-growing techniques for more accurate and efficient point cloud clustering.

Awards, Skills and Hobbies

- **2024 CVPR-Building 3D Challenge Winner**, Team Leader, Led the team to win first place in this challenge focused on constructing wireframe models for Building3D roof point cloud datasets, implementing advanced point cloud feature extraction and clustering algorithms.
- **Consistently awarded university scholarships and honors such as "Outstanding Student Cadre"**.

- **International Mathematical Contest in Modeling (MCM), Honor Award.**
- **Acquired extensive practical experience and skills.** Building algorithm visualization platforms (e.g., Gradio); Developing on large-scale GPU clusters; Constructing and fine-tuning Multimodal AI agents.
- **Hobbies.** Enjoying Hiking, Gym, Basketball.